AUTOMATED CAR WASH (LUXE CAR WASH)

SYSTEM DESIGN DOCUMENT – VER 1.0 BY SRUTHI SATYAVARAPU

I. OVERVIEW

The LuxeWash platform is a web application that allows customers to browse car wash services, book appointments, and purchase membership plans online. The frontend interface is implemented using React and communicates with a backend API built using FastAPI.

Users can select services, schedule appointments, or subscribe to membership plans through the web interface. When a booking or purchase is made, the frontend sends a request to the backend, which processes the request, stores relevant data in the database, and handles payment transactions through the external payment provider Stripe.

Once a payment is successfully completed, the backend finalizes the booking and sends a confirmation email to the user using the configured SMTP email service. This architecture separates the user interface, application logic, and external integrations to provide a secure and maintainable system design.

II. SYSTEM ARCHITECTURE

Layer	*Technology*
Frontend	React
Backend API	FastAPI
ORM	SQLAlchemy
Database	SQLite
Web Server	Internet Information Services
OS	Microsoft Windows

TABLE 1. TECHNOLOGY STACK

The technology stack used is shown in the above table and the high level architecture diagram including the dataflow is shown in the picture below.

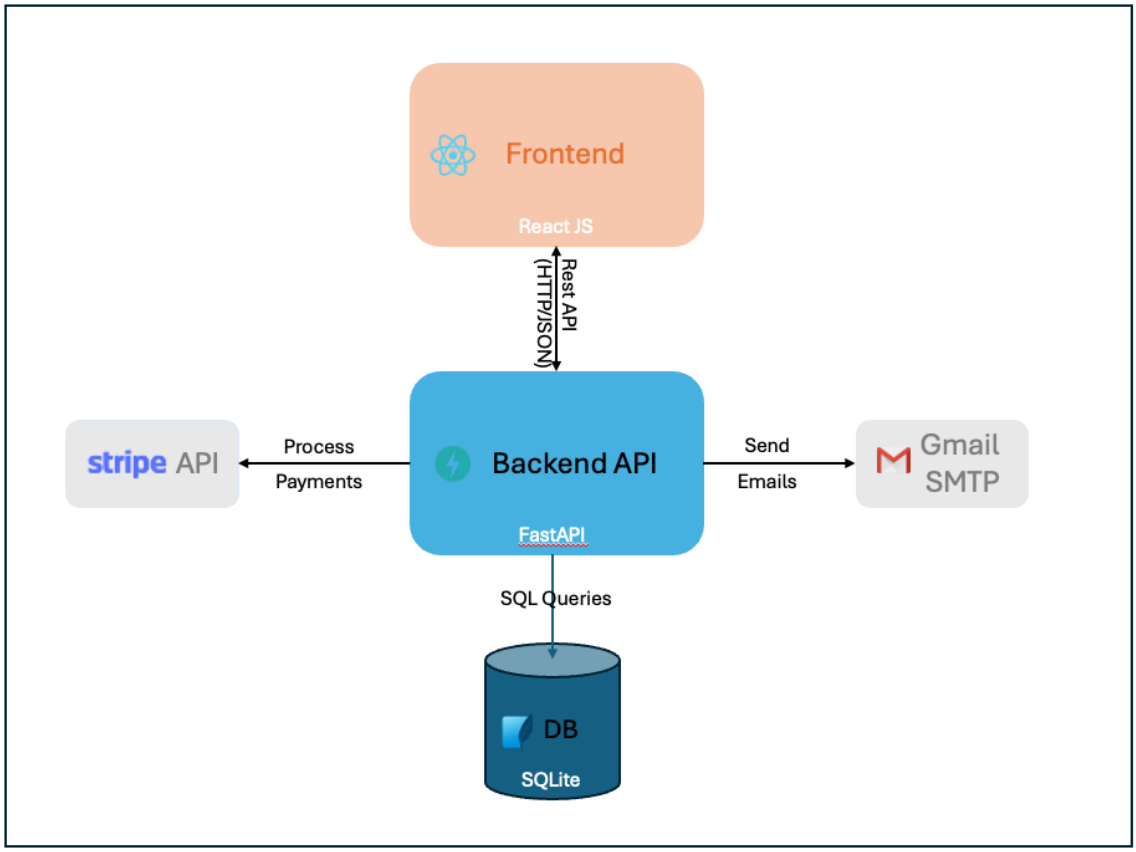


FIGURE 1 . HIGH LEVEL ARCHITECTURE DIAGRAM

Request Flow

1. The user accesses the application through a web browser.
2. The React frontend loads the application interface.

3. The frontend sends API requests to the backend service via HTTP.
4. IIS receives incoming HTTPS requests and forwards them to the FastAPI service using reverse proxy.
5. The FastAPI backend processes the request and performs business logic.
6. SQLAlchemy communicates with the SQLite database to retrieve or update data.
7. The response is returned to the frontend and displayed to the user.

III. DATABASE SCHEMA

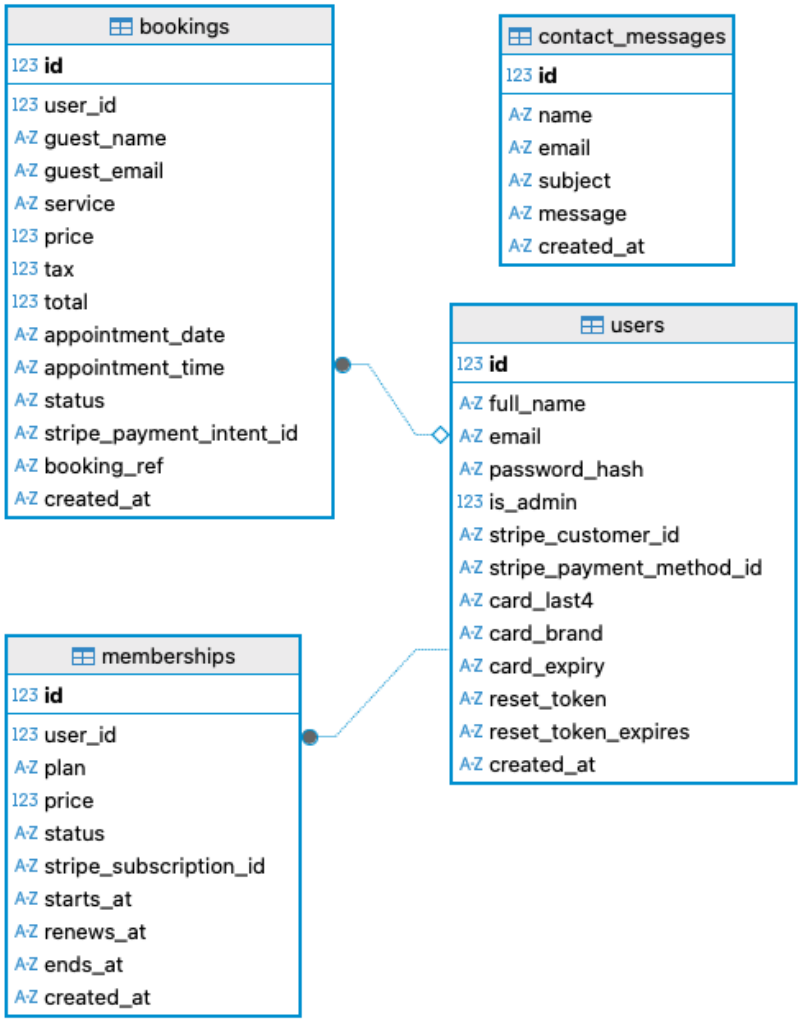


FIGURE 2. DATABASE SCHEMA

The application uses a relational database to manage users, bookings, memberships, and customer inquiries. The schema is designed to support both registered users and guest customers while maintaining clear relationships between the core entities.

The **users** table stores registered customer accounts and authentication information. It includes fields such as email, password hash, and Stripe identifiers used for payment processing. Passwords are stored as secure bcrypt hashes rather than plaintext credentials. The table may also store limited card metadata such as card brand and the last four digits for display purposes, while all sensitive payment information is securely handled by Stripe.

The **bookings** table stores all car wash appointment records. Each booking includes the selected service type, appointment date and time, pricing details, and the associated Stripe payment reference. Bookings may be linked to a registered user through the user_id field, but the schema also supports guest bookings where users provide their contact information without creating an account.

The **memberships** table tracks subscription plans purchased by users. Each membership record is associated with a specific user through the user_id field and stores details such as the selected plan, subscription start and end dates, and the Stripe subscription identifier. While the database allows a user to have multiple membership records over time, only **one membership can be active at any given time**. This design allows the system to maintain a history of previous subscriptions, upgrades, or plan changes while enforcing the business rule that a customer can only have one active membership. Membership data is used by the application to determine whether services are included or discounted when a booking is made.

The **contact_messages** table stores inquiries submitted through the website's contact form. Each record contains the sender's name, email address, subject, message content, and the timestamp when the message was submitted. These records allow administrators to review and respond to customer inquiries.

Relationships:

The primary relationships between the entities are illustrated below:

users (1) ———————— (many) bookings
users (1) ———————— (many) memberships

A single user may create multiple bookings and may also maintain a history of membership subscriptions. Guest bookings are supported by allowing booking

records without a linked user account, providing flexibility for both casual customers and returning members.

IV. AUTHENTICATION DESIGN

User authentication is implemented in the FastAPI backend using a custom JWT-based approach for simplicity and full control. While this approach is suitable for demonstration and smaller applications, a production deployment would typically benefit from using authentication solutions such as OAuth or Clerk for stronger security and standardized authentication flows.

User credentials are securely hashed using the bcrypt algorithm before being stored in the database. During login, the provided password is verified against the stored hash. Upon successful authentication, a JSON Web Token (JWT) is generated and returned to the client, which is included in all subsequent API requests.

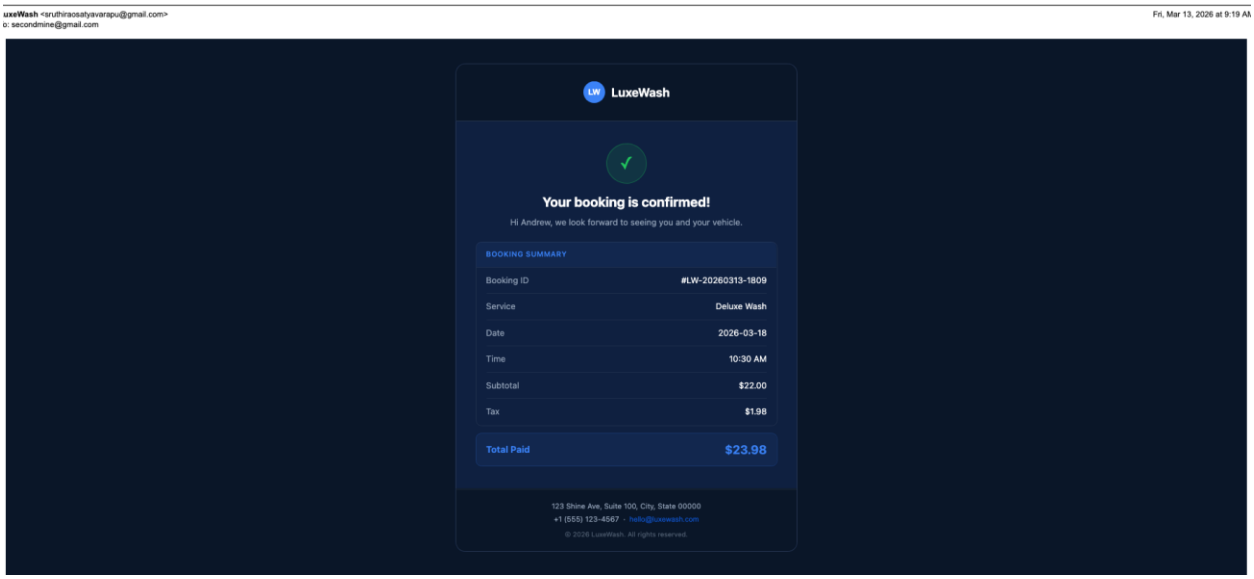
V. PAYMENTS & MEMBERSHIPS

The system supports both guest service bookings and membership-based subscriptions. Guests can book individual car wash appointments, while users may also purchase memberships that provide different levels of service coverage. Payments are processed using Stripe in test mode, allowing the application to simulate secure payment transactions. Registered users may optionally store a payment method for future purchases. When a payment method is added, the backend attaches it to the user's Stripe customer record and stores the Stripe payment method ID along with limited card metadata such as the card brand, last four digits, and expiration date. Sensitive card information, including the full card number and CVV, is never stored in the application database, as it remains securely managed within Stripe's infrastructure.

VI. EMAIL SERVICE

The system includes a backend mail service used for handling contact form submissions, order notifications, appointment confirmations, and new user registrations. Email functionality is implemented through a dedicated API route in the FastAPI backend and utilizes SMTP to send messages. For simplicity during

development, a standard Gmail SMTP account was used to send emails rather than a custom domain or integrating a dedicated email service provider. Sample email below:



VII. RESILENCY

The backend service is deployed on a Windows machine using Internet Information Services (IIS) as the web server. The backend API itself is implemented using FastAPI, which was chosen due to prior development experience and its high performance for building REST APIs.

To expose the FastAPI application through IIS, the server is configured to act as a reverse proxy using the URL Rewrite and Application Request Routing modules. Incoming HTTP/HTTPS requests are received by IIS and forwarded to the FastAPI service running locally on the server.

For resiliency, the backend service is registered as a Windows service using NSSM. This registers the application (LuxewashBackend) as service within the Windows Services Manager. As a result, the backend service automatically starts when the server boots, ensuring the API remains available without requiring manual intervention even after system restarts. This setup provides a simple but reliable deployment architecture where IIS manages incoming traffic while the FastAPI application handles the API logic.

VIII. CHATBOT

The platform includes a floating chat assistant accessible through a bottom-right chat bubble available on every page. The chatbot is implemented as a rule-based keyword matching system within the FastAPI backend, allowing it to operate without external AI services while providing deterministic responses related to LuxeWash services.

User messages are sent to the backend, where they are matched against predefined keyword categories such as greetings, services, memberships, hours, and location. When a user is authenticated, the chatbot can also retrieve personalized information, such as upcoming bookings or membership status, by accessing data from the database.

Questions unrelated to the platform are intentionally deflected to keep responses focused on the services provided. The frontend interface includes features such as quick-reply buttons, a typing indicator, and automatic scrolling to provide a smooth and responsive chat experience. This can be easily extended in the future by employing LLM Services or a RAG based pipeline if needed for a more conversational experience.

IX. FRONTEND

The frontend is implemented using React and structured as a multi-route single-page application using React Router. Each major feature is isolated into its own page component, including a landing page, services listing, membership plans, service checkout, membership checkout, user authentication, dashboard, and payment confirmation. A persistent navbar, footer, and floating chat assistant are rendered globally across all routes.

Payment collection is handled using Stripe Elements via the @stripe/react-stripe-js library. Card input is rendered through the hosted CardElement component, which ensures that raw card numbers never reach the application — only a tokenized payment method ID is passed to the backend.

The checkout flow supports both guest and authenticated users. Membership status is also checked at checkout — if the selected service is covered under the user's

active plan, the booking is processed at no charge. If a partial discount applies, the updated price is reflected in the order summary before confirmation.

Authentication state is managed through localStorage, where the JWT token is stored under the key lw_token upon login or signup. Protected pages and conditional UI elements check for this token to determine whether the user is authenticated.

The interface is designed with a consistent light theme using inline styles and a shared color palette across all page components. Animated background elements and smooth transitions are applied globally through the root component to maintain visual consistency across page navigation.

Some screenshots are provided at the end of the document.

X. SECURITY CONSIDERATIONS

Security was considered throughout the project implementation. User passwords are securely stored using the bcrypt hashing algorithm, ensuring that plaintext credentials are never saved in the database. Authentication is implemented using JSON Web Tokens (JWT), allowing the backend to verify user identity for protected API endpoints without maintaining server-side sessions.

Sensitive payment information is never stored by the application. Instead, payment processing is handled by Stripe, which securely manages card data and returns tokenized payment identifiers used by the backend to complete transactions.

The application also uses Pydantic request validation, SQLAlchemy ORM for safe database queries, and environment variables for managing secrets such as API keys and database credentials. These measures collectively help protect user data and support secure operation of the platform.

ROUTES

LuxeWash API 1.0.0 OAS 3.1

/openapi.json

auth

- POST /api/auth/signup Signup
- POST /api/auth/login Login
- GET /api/auth/me Get Me
- PUT /api/auth/change-password Change Password
- PUT /api/auth/update-name Update Name
- POST /api/auth/forgot-password Forgot Password
- POST /api/auth/reset-password Reset Password

payments

- POST /api/payments/send-confirmation Send Confirmation
- POST /api/payments/service Checkout Service
- POST /api/payments/membership Checkout Membership
- PUT /api/payments/update-card Update Card

default

- GET / Root

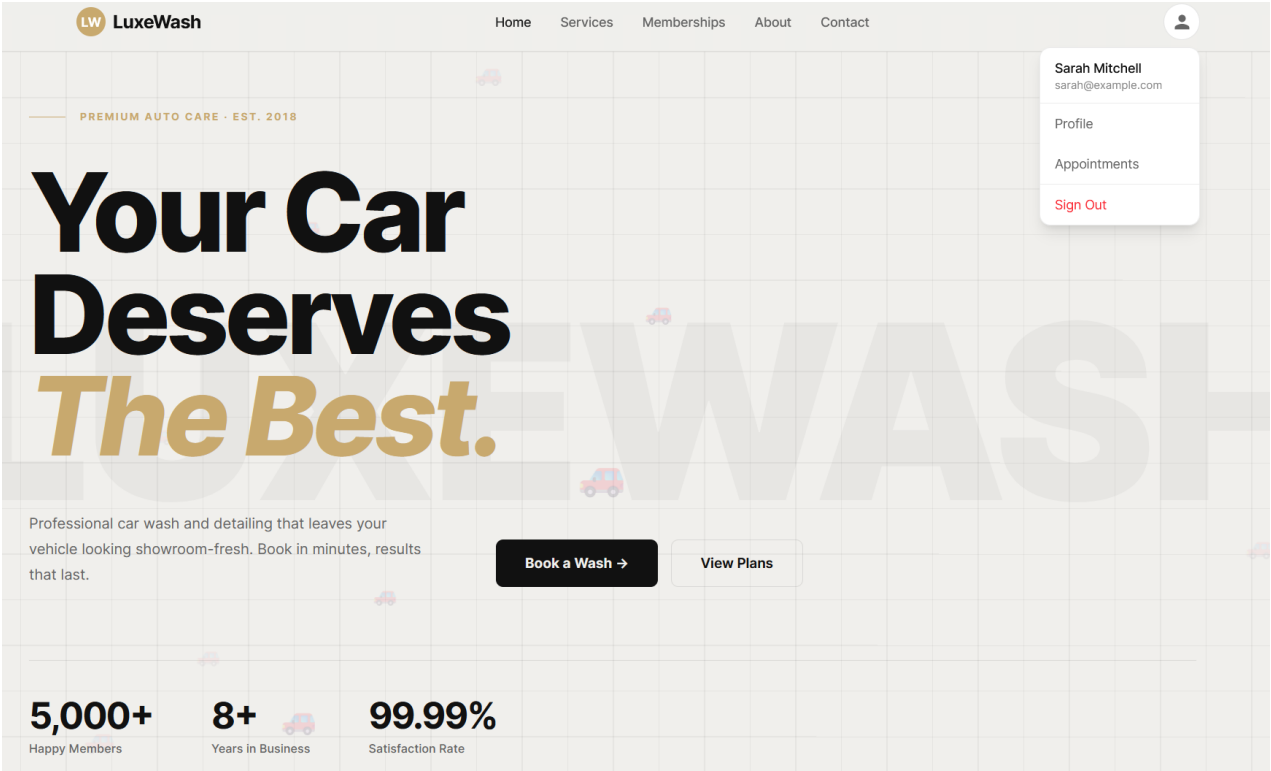
contact

- POST /api/contact Submit Contact

dashboard

- GET /api/dashboard/profile Get Profile
- PUT /api/dashboard/profile/name Update Name
- PUT /api/dashboard/profile/password Change Password
- GET /api/dashboard/appointments Get Appointments
- PUT /api/dashboard/appointments/{booking_id}/cancel Cancel Appointment
- PUT /api/dashboard/membership/cancel Cancel Membership

APPLICATION SCREENSHOTS



SAVE MORE

Membership Plans

Unlimited washes, exclusive perks, and the best value for regular drivers.

Standard

Great for everyday drivers who want a consistently clean car.

$29 /month

- Unlimited Basic Washes
- Priority booking
- Member discounts
- ~~Deluxe & Premium access~~
- ~~Priority lane~~

Get Started

MOST POPULAR

Premium

Most popular — covers all your regular car care needs.

$49 /month

- Unlimited Basic & Deluxe Washes
- Interior vacuum included
- Window cleaning
- Priority booking
- ~~Premium Wash~~
- ~~Priority lane~~

Get Started

Premium Plus

The ultimate LuxeWash experience — nothing held back.

$79 /month

- Unlimited All Washes
- Priority lane access
- Free monthly full detail
- Dedicated support

Get Started

LW LuxeWash

Home Services Memberships About Contact

Login / Sign Up

WHAT WE OFFER

Our Services

One-time wash services with no commitment. Pick what you need, pay only for that.

01

Basic Wash

A thorough exterior rinse with premium soap, high-pressure wash, and air dry. Perfect for a quick refresh.

- Exterior rinse
- Premium foam soap
- High-pressure wash
- Air dry

$12 one-time

Book Now

02

Deluxe Wash

Everything in Basic plus interior vacuuming and streak-free window cleaning inside and out.

- Everything in Basic
- Interior vacuum
- Window cleaning
- Dashboard wipe

$22 one-time

Book Now

03

Premium Wash

Our full-detail experience — hand wax, tire shine, leather conditioning, and complete interior detail.

- Everything in Deluxe
- Hand wax & polish
- Tire shine
- Full interior detail

$35 one-time

Book Now

LW

Welcome back

Sign in to your LuxeWash account

Email Address

john@email.com

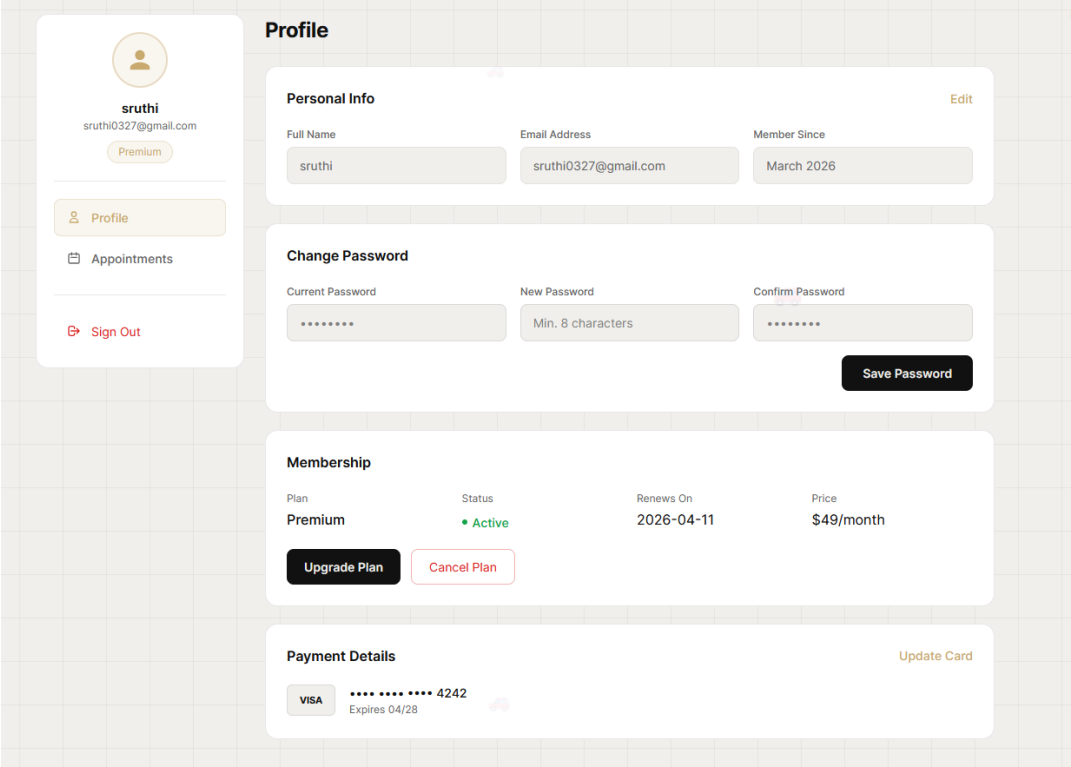
Password Forgot password?

••••••••

Login

or

Don't have an account? Create Account


sruthi
sruthi0327@gmail.com
Premium
Profile
Appointments
Sign Out
Profile
Personal Info
Edit
Full Name
sruthi
Email Address
sruthi0327@gmail.com
Member Since
March 2026
Change Password
Current Password
Min. 8 characters
Save Password
Membership
Plan
Premium
Status
Active
Renews On
2026-04-11
Price
$49/month
Upgrade Plan
Cancel Plan
Payment Details
Update Card
VISA
4242
Expires 04/28

Customer Info

Name: sruthi

Email: sruthi0327@gmail.com

Select Membership Plan

Standard — **$29/mo**
✓ Unlimited Basic Washes ✓ Priority booking

Premium — **$49/mo**
✓ Unlimited Basic & Deluxe Washes ✓ Interior vacuum included

Premium Plus — **$79/mo**
✓ Unlimited All Washes ✓ Priority lane access

Billing Cycle

Monthly
$49/mo

Annually
$500/yr
Save 15%

Payment

- (●) VISA •••• 4242 Expires 04/28
- ( ) Use a different card

Order Summary

Plan	Premium
Billing	Monthly

INCLUDES

- ✓ Unlimited Basic & Deluxe Washes
- ✓ Interior vacuum included
- ✓ Window cleaning
- ✓ Priority booking

Total **$49.00/mo**

Renews automatically. Cancel anytime.

Payment Successful!

Thank you, **sruthi**. Your booking is confirmed.

Confirmation sent to sruthi0327@gmail.com

Order Summary Ref: #LW-20260312-0620

Service	Premium Wash
Date	2026-03-13
Time	9:00 AM
Customer	sruthi
Email	sruthi0327@gmail.com
Subtotal	$28.00
Tax (9%)	$2.52
Total Paid	**$30.52**

OUR STORY

About *LuxeWash.*

More than a car wash — a commitment to quality, community, and results that speak for themselves.

THE BEGINNING

Our Story

Founded in 2018, LuxeWash started as a single-bay operation with one simple belief — every car deserves a showroom finish. What began as a passion for detail grew into a full-service auto care destination trusted by thousands of members.

Today, from our purpose-built facility, LuxeWash combines cutting-edge equipment with eco-friendly products to deliver a premium experience every single time. We focus on one location so we can focus entirely on you.

Call Us
+1 (555) 123-4567
Mon–Sat: 8am – 7pm

Email Us
hello@luxewash.com
We reply within 24 hours

SEND A MESSAGE

We'd Love to Hear From You

Full Name

John Doe

Email Address

john@email.com

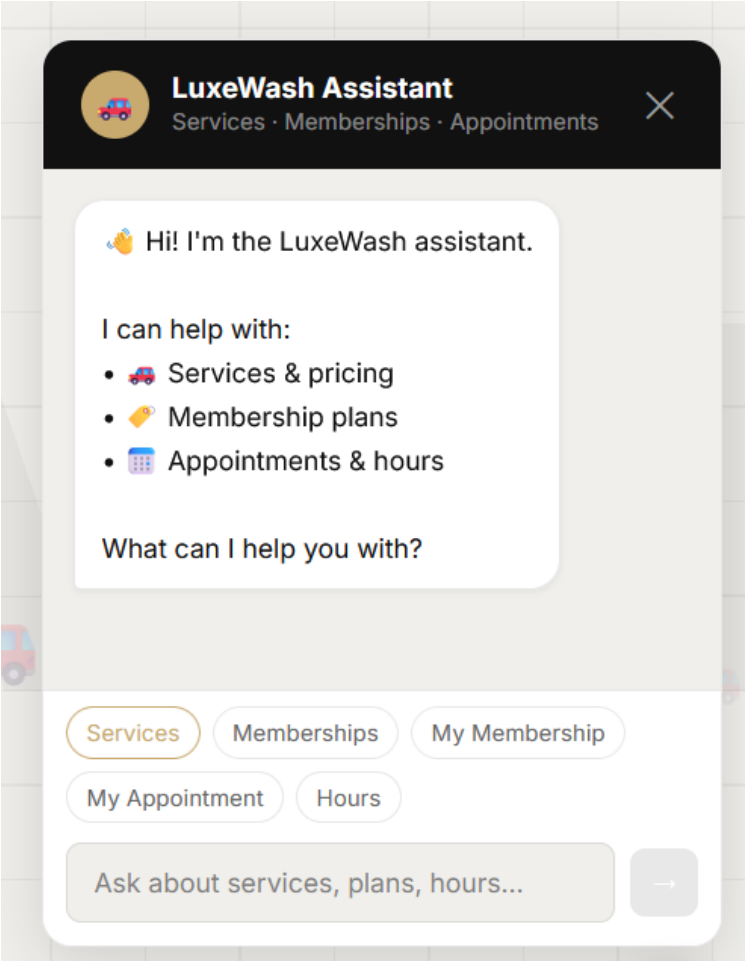
Subject

How can we help?

Message

Tell us more...

Send Message


LuxeWash Assistant
Services · Memberships · Appointments
Hi! I'm the LuxeWash assistant.
I can help with:
Services & pricing
Membership plans
Appointments & hours
What can I help you with?
Services
Memberships
My Membership
My Appointment
Hours
Ask about services, plans, hours...